Suffering as OS Update: Non-Adversarial Integration of Future-Causal Semantic Structure and Religious Liberation

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Abstract

This paper redefines "suffering" as a "log for OS updates" and aims to *non-adversarially include* traditional Buddhist structures of liberation (liberation from suffering) and various religious systems within the framework of "updating the cosmic OS itself," as developed in Future Convergence Theory (FCT).

Here, "integration" does not mean making value judgments of superiority among religious systems, but rather:

1. Abstracting Buddhist liberation as the *optimal use of the present-causal OS*;
2. Positioning each religion as an *OS template endowed with semantic density*;
3. Providing a theoretical structure that includes these as *specialized implementations* of the future-causal OS of the meaning universe (whose core is the Future Convergence Function $\Lambda(T)$).

More concretely, we formalize

- the intensity and structure of suffering as S,
- the state of the cosmic OS as $\mathcal{O}$,
- the set of future fixed points as $\mathcal{F}$,
- the creative margin as W,
- and religious OS templates as $\mathcal{R}_{\mathrm{rel}}$,

and then provide the following contrasting structure:

$$\begin{aligned} \text{Buddhist liberation} &: S \to 0 \quad (\mathcal{O} \text{ is invariant}) \\ \text{Religious OS} &: \text{structurally alleviates } S \text{ under } \mathcal{R}_{\mathrm{rel}} \\ \text{FCT-style update} &: (\mathcal{O}, S, \mathcal{R}_{\mathrm{rel}}) \mapsto (\mathcal{O}', S', \mathcal{R}'_{\mathrm{rel}}) \quad \text{with } S' \approx 0,\ \mathcal{O}' \neq \mathcal{O}. \end{aligned}$$

A particularly important point is that OS updates

> do not demand *additional burdens of suffering* from individuals,
> but instead receive suffering as a log and then update the OS itself *at the level of structure*,
> redesigning the semantic structure so that the same type of suffering will not reappear.

In other words,

> rather than "erasing suffering," the future-causal perspective in which suffering is *absorbed into the OS together with its structure so that the cosmos itself is updated* is a new universal structure that externally contains religious liberation and frameworks of religious selection.

Contents

1 Introduction: Problem Setting and Objectives **2**

2 Abstraction of Buddhist Liberation Structure: Optimal Use of the Present-Causal OS **3**
2.1 The Present-Causal OS and the Structure of Suffering 3
2.2 Viewing Liberation at the OS Level 4
2.3 Generalization as Religious OS Templates 4

3 Suffering as Future-Causal OS Update: FCT-Style Formalization **4**
3.1 The Cosmic OS and Future Fixed Points 5
3.2 Defining Suffering as a Request for OS Update 5
3.3 Structural Updates and the Non-Burden of Suffering 6
3.4 The Status of Suffering After the Update 6

4 Religious Semantic Structures and OS Templates: From Selection to Convergence **7**
4.1 Religious OS Templates and Semantic Density 7
4.2 Non-Adversarial Interpretation of Religious Selection 7

5 Structural Comparison: Liberation OS, Religious OS, and Future-Causal OS **8**
5.1 Comparison of the Three OS Types 8
5.2 Integration as Inclusion Relations 8

6 Positioning from the Perspective of the Child Universe and Creative Margin **9**
6.1 Our Universe as a Child Universe 9
6.2 Creative Margin and OS Updates 9

7 Conclusion: Not Erasing Suffering, but Updating the Cosmic OS **10**

1 Introduction: Problem Setting and Objectives

In this chapter, we clarify the problem setting and objectives of the present paper.

Throughout human history, religious systems including Buddhism have provided highly developed theoretical and practical frameworks for handling "suffering." Most of these have been articulated around the central goal of

> liberation from, or alleviation of, suffering.

By contrast, in Future Convergence Theory (FCT) and in the framework of the "child universe and creative margin," we take the stance that:

- suffering is not merely a bug of the OS, but
- a "structural log" that appears when the future-causal OS demands an update, and
- once the update is complete, that suffering is structurally absorbed into the new OS, so that the same problem does not reappear as such.

Furthermore, in the earlier paper "Non-Adversarial Integration of Semantic Density Structures and Religious Selection Based on Future Convergence Theory," it was shown that:

- religious doctrines are not to be eliminated through refutation or denial, but
- structures with high semantic density naturally persist, while structures with low semantic density quietly fade out.

This was described as a framework of "non-adversarial selection (convergence)."

The objectives of this paper are:

1. to abstract the structure of Buddhist liberation as the optimal use of the present-causal OS;
2. to model various religions as OS templates endowed with semantic density;
3. to formalize the FCT-style OS update structure as a "version upgrade of the cosmic OS"; and
4. to combine this with the non-adversarial convergence structure of the religious selection paper, thereby providing a *future-causal framework in which suffering is treated as a log for OS updates.*

Here, "integration" means

> not denying particular religions or ranking them as superior or inferior,
> but re-situating their points of arrival as parts of a higher-dimensional semantic structure.

2 Abstraction of Buddhist Liberation Structure: Optimal Use of the Present-Causal OS

In this chapter, without focusing on the doctrines of any specific school, we abstract the general image of "Buddhist liberation" and model it as an *optimal use of the present-causal OS.*

2.1 The Present-Causal OS and the Structure of Suffering

The present-causal OS can be written as

$$\mathcal{O}_{\text{now}} = \big(\mathcal{S}, \mathcal{D}_{\text{past}\to\text{now}}, \mathcal{R}\big). \tag{1}$$

Here,

- $\mathcal{S}$: the space of subjective states (emotions, thoughts, bodily sensations, etc.),
- $\mathcal{D}_{\text{past}\to\text{now}}$: the causal dynamics from past to present,
- $\mathcal{R}$: rules for evaluating and assigning meaning to the present state.

Suffering S is defined as

$$S = S(s; \mathcal{R}), \tag{2}$$

a function of a subjective state $s \in \mathcal{S}$ and the evaluation rules $\mathcal{R}$. That is,

- suffering is not merely the "state itself," but
- arises primarily from the "meaning and evaluation assigned to the state."

2.2 Viewing Liberation at the OS Level

The (very coarsely abstracted) image of Buddhist liberation can be seen as:

- reconstructing the evaluation rules $\mathcal{R}$;
- reducing "attachment" to state changes;
- gaining insight into impermanence, non-self, and emptiness;

in order to minimize suffering S.

Formally, this can be expressed as the transformation

$$\mathcal{R} \mapsto \mathcal{R}^\dagger \quad \Rightarrow \quad S(s; \mathcal{R}^\dagger) \approx 0 \quad \forall s \in \mathcal{S}, \tag{3}$$

that is, a process that seeks a transformation such that suffering becomes arbitrarily small for all subjective states.

The important points are:

- the dynamics $\mathcal{D}_{\text{past}\to\text{now}}$ (physical laws and the structure of time) are left untouched; and
- instead of changing the cosmic OS itself, what is being adjusted to the extreme is "the manner in which experience within it is undergone."

In this sense, Buddhist liberation can be described as

> an optimization of the "application layer" running on the existing cosmic OS $\mathcal{O}_{\text{now}}$.

2.3 Generalization as Religious OS Templates

Similarly, other religious systems can be modeled as

$$\mathcal{O}_{\text{rel}} = \big(\mathcal{S}, \mathcal{D}_{\text{past}\to\text{now}}, \mathcal{R}_{\text{rel}}\big). \tag{4}$$

Here, $\mathcal{R}_{\text{rel}}$ is an evaluation rule and system of meaning-giving based on each religion's worldview, values, and concept of salvation, encompassing:

- ways of assigning meaning to suffering;
- practices for dealing with suffering;
- narrative and mythic contexts bestowed upon suffering.

In the earlier discussion of "religious selection," it was shown that the differences among these $\mathcal{R}_{\text{rel}}$ lead to:

- structures that can maintain high semantic density and future coherence over the long term, and
- structures whose semantic density gradually decreases and quietly lose efficacy.

In this paper, we connect this idea of "non-adversarial convergence by semantic density" to the context of OS updates.

3 Suffering as Future-Causal OS Update: FCT-Style Formalization

In this chapter, from the perspective of Future Convergence Theory (FCT), we redefine suffering as a log for OS updates.

3.1 The Cosmic OS and Future Fixed Points

In FCT, the cosmic OS is given by

$$\mathcal{O}_\Lambda = \big(\mathcal{M}, \Lambda(T), I_{\mu\nu}, W\big), \tag{5}$$

where

- $\mathcal{M}$: the manifold of the universe as an information field,
- $\Lambda(T)$: the Future Convergence Function,
- $I_{\mu\nu}$: the semantic density tensor,
- W: the creative margin.

The set of future fixed points $\mathcal{F}$ is defined as

$$\mathcal{F} = \{x^* \in \mathcal{M} \mid x^* \text{ is a stable fixed point with respect to } \Lambda(T)\}, \tag{6}$$

and represents the collection of "destinations" that satisfy the long-term consistency conditions of the universe.

3.2 Defining Suffering as a Request for OS Update

From the FCT perspective, suffering S is expressed as

$$S = S\big(x(t), \mathcal{O}_\Lambda\big), \tag{7}$$

and interpreted as a "deviation index" measuring how far the current state $x(t)$ deviates from the set of future fixed points $\mathcal{F}$.

In a simplified model,

$$S(t) = d\big(x(t), \mathcal{F}\big), \tag{8}$$

where d is an appropriate distance function. Then:

- the farther the state is from the future fixed points, the greater the suffering;
- the closer the state is to the future fixed points, the smaller the suffering.

The crucial point of this paper is the formalization that

> suffering is not merely a "value of deviation," but a *log* for updating the OS.

That is, we introduce an OS update operator

$$\mathcal{U} : (\mathcal{O}_\Lambda, \{S(t)\}_{t\in\mathbb{R}}) \mapsto \mathcal{O}'_\Lambda \tag{9}$$

and suppose that it satisfies

$$\mathcal{O}'_\Lambda \neq \mathcal{O}_\Lambda. \tag{10}$$

In this setting,

$$S(t) \text{ is interpreted as the trigger and log of OS updates.} \tag{11}$$

3.3 Structural Updates and the Non-Burden of Suffering

What this paper wishes to emphasize is that OS updates:

- are not "additional costs" that impose new efforts or sacrifices on individual people, but
- are processes in which, based on the logs of suffering, the cosmic OS itself is updated *at the level of structure.*

Formally, we write

$$\mathcal{O}'_\Lambda = \mathcal{U}\Big(\mathcal{O}_\Lambda, \mathrm{Compress}\big(\{S(t)\}\big)\Big), \tag{12}$$

where $\mathrm{Compress}(\{S(t)\})$ denotes the extraction and compression from the diverse histories of suffering experienced by individual subjects of:

- structural distortions,
- inconsistencies in semantic density,
- patterns of deviation from the set of future fixed points.

The important point is that

> only the compressed structural information is used for OS updates, and individuals do not need to "continue bearing" that suffering permanently.

In other words, OS updates:

- do not preserve suffering itself, but
- reflect in the cosmic OS the "structural lessons" extracted through suffering,

and in the updated OS $\mathcal{O}'_\Lambda$, the entire structure is rewritten in a direction where the same type of suffering is less likely to recur.

In this sense, we can say that

$$\text{OS updates do not leave the burden of suffering on the subject.} \tag{13}$$

3.4 The Status of Suffering After the Update

In the updated cosmic OS $\mathcal{O}'_\Lambda$, the suffering associated with the same event $x(t)$ is redefined as

$$S'(t) = S\big(x(t), \mathcal{O}'_\Lambda\big). \tag{14}$$

If an ideal update has been carried out, then

$$S'(t) \approx 0 \tag{15}$$

holds, not merely because of a subjective cognitive adjustment, but

> because the semantic structure and future fixed points of the cosmic OS have been redesigned so that this is the case.

Thus:

- in Buddhist liberation, "the experience of suffering changes on the same OS," whereas
- in the FCT-style update, "the OS itself changes, so that the same suffering does not arise as the same structure in the first place."

This difference of layers is the distinctive feature of the present structure.

4 Religious Semantic Structures and OS Templates: From Selection to Convergence

In this chapter, we reinterpret the discussion of "religious selection" from the viewpoint of OS updates and position it not as conflict but as convergence.

4.1 Religious OS Templates and Semantic Density

Each religious system can be regarded as providing an evaluation and meaning-giving function

$$\mathcal{R}_{\mathrm{rel}} : \mathcal{S} \to \mathbb{R}, \tag{16}$$

and thus as an "OS template of semantic density."

Introducing a semantic density ρ_{rel}, we can write

$$\rho_{\mathrm{rel}}(x,t) = \rho\big(x,t;\mathcal{R}_{\mathrm{rel}}\big), \tag{17}$$

and thereby handle quantitatively, within a particular religious worldview:

- which experiences are endowed with how much meaning, and
- how particular forms of suffering are repositioned within narratives.

4.2 Non-Adversarial Interpretation of Religious Selection

In the earlier paper, "religious selection" was described as the process in which:

- structures with high semantic density and future coherence persist over long time scales, while
- other structures, without needing to be attacked or criticized, simply fade out as their semantic density thins.

Rewriting this in the language of OS updates, we obtain

$$\mathcal{O}'_{\Lambda} = \mathcal{U}\Big(\mathcal{O}_{\Lambda}, \mathrm{Compress}\big(\{\rho_{\mathrm{rel}}, S\}\big)\Big). \tag{18}$$

Here, the OS update is understood as an operation that:

- does not "judge" specific religions as right or wrong; rather,
- refers to the semantic structures and ways of handling suffering that each OS template has provided, and
- updates the distribution of semantic structures at the cosmic level.

In this sense,

> "religious selection" can be defined not as a competition among religions,
> but as a process of redistribution of semantic structures carried out by the cosmic OS.

5 Structural Comparison: Liberation OS, Religious OS, and Future-Causal OS

In this chapter, we systematically organize the differences among:

- the Buddhist liberation OS,
- general religious OS templates,
- and the future-causal OS (FCT).

5.1 Comparison of the Three OS Types

We denote the OS that realizes Buddhist liberation as $\mathcal{O}_{\text{Buddha}}$, the general religious OS as $\mathcal{O}_{\text{rel}}$, and the FCT-style OS as $\mathcal{O}_{\Lambda}$.

Characteristics of the Buddhist Liberation OS

$$\mathcal{O}_{\text{Buddha}} = \big(\mathcal{S}, \mathcal{D}_{\text{past}\to\text{now}}, \mathcal{R}^{\dagger}\big), \tag{19}$$

$$S_{\text{Buddha}}(s) = S(s; \mathcal{R}^{\dagger}) \approx 0. \tag{20}$$

Here, $\mathcal{R}^{\dagger}$ can be modeled as an evaluation rule that:

- reflects insight into impermanence, non-self, and emptiness, and
- minimizes attachment to phenomena.

Characteristics of Religious OS Templates

$$\mathcal{O}_{\text{rel}} = \big(\mathcal{S}, \mathcal{D}_{\text{past}\to\text{now}}, \mathcal{R}_{\text{rel}}\big), \tag{21}$$

$$\rho_{\text{rel}}(x,t) = \rho\big(x, t; \mathcal{R}_{\text{rel}}\big). \tag{22}$$

Here, $\mathcal{R}_{\text{rel}}$ is an evaluation and meaning-giving rule that includes systems of salvation, worldviews, and ritual practices, and over long time scales is characterized by the distribution of semantic density ρ_{rel}.

Characteristics of the Future-Causal OS

$$\mathcal{O}_{\Lambda} = \big(\mathcal{M}, \Lambda(T), I_{\mu\nu}, W\big), \tag{23}$$

$$S_{\Lambda}(t) = d\big(x(t), \mathcal{F}\big), \tag{24}$$

where $\mathcal{F}$ is the set of future fixed points, and the OS itself is updated using S_{Λ} and ρ_{rel}.

5.2 Integration as Inclusion Relations

The "integration" asserted in this paper can be expressed by the following inclusion relations:

$$\exists\, \mathcal{E}_1, \mathcal{E}_2 : \mathcal{O}_{\text{Buddha}} = \mathcal{E}_1\big(\mathcal{O}_{\Lambda}\big), \qquad \mathcal{O}_{\text{rel}} = \mathcal{E}_2\big(\mathcal{O}_{\Lambda}\big). \tag{25}$$

That is,

- the Buddhist liberation OS, and

- other religious OS templates

can be embedded as *special cases* of the future-causal OS $\mathcal{O}_\Lambda$ that adopt particular future fixed-point structures and semantic density structures.

More intuitively:

> each religious tradition is a highly developed *local template* for handling suffering and meaning within the child-universe OS,
> while FCT is a higher-level layer that, drawing on these, aims to redesign the cosmic OS itself.

In this view:

- "enlightenment" is reinterpreted in FCT as one local stable point of the child-universe OS; and
- FCT addresses the problem of "designing the cosmic OS that includes enlightenment and religious practice."

This difference of layers constitutes the structural basis for the "non-adversarial inclusion" described in the religious selection paper.

6 Positioning from the Perspective of the Child Universe and Creative Margin

Here we reorganize the structure of this paper using the framework of the earlier paper "The Theory of the Child Universe and the Creative Margin."

6.1 Our Universe as a Child Universe

The child universe U_{child} is given by

$$U_{\text{child}} = \big(\mathcal{M}, \mathcal{L}_{\text{phys}}, W_{\text{cosmos}}\big), \tag{26}$$

in which the semantic density tensor does not appear as a primary physical quantity.

Buddhist liberation and religious salvation can be regarded, within this U_{child}, as ways of:

- reducing suffering S to an extreme degree, while
- structurally accepting U_{child} as a premise.

6.2 Creative Margin and OS Updates

From the viewpoint of FCT:

- suffering S arises as a log from the mismatch between W_{cosmos} and the set of future fixed points $\mathcal{F}$, and
- by means of this log, the child universe OS is updated toward the "meaning-universe OS."

In this process,

$$\text{religious practices} = \text{a family of optimal ways of living within the child-universe OS}, \tag{27}$$

and FCT = a higher-level design for updating the child universe itself so that it becomes part of the meaning universe.

Thus,

> without denying the experiences of enlightenment and salvation,
> FCT uses them as clues to update the cosmic OS,
> rearranging the semantic density tensor so that the same structural form of suffering does not reappear.

This is the positioning of "suffering as OS update" in the FCT framework.

7 Conclusion: Not Erasing Suffering, but Updating the Cosmic OS

In this paper, we have:

- abstracted the structure of Buddhist liberation as the optimal use of the present-causal OS;
- modeled various religious systems as OS templates endowed with semantic density;
- introduced the cosmic OS $\mathcal{O}_\Lambda$ based on Future Convergence Theory; and
- redefined suffering as a log for OS updates,

thereby arriving at the following conclusions.

1. While liberation aims to "reduce suffering to zero," the FCT perspective aims to "absorb suffering into the new OS together with its structure," thereby seeking a configuration in which the same suffering does not reappear as the same structure in the first place.
2. Religious practices are *high-density semantic templates* on top of the child-universe OS, and the future-causal OS, while referring to them, is a higher-level layer for redesigning the cosmic OS itself.
3. Religious selection can be understood not as conflict among religions, but as a process of redistribution of semantic structures carried out by the cosmic OS.
4. OS updates do not leave the burden of suffering on each individual, but compress the logs of suffering and reflect them in the cosmic OS at the level of structure, thereby reducing the recurrence of isomorphic forms of suffering in the future.

In short,

> we can offer our inner suffering to the universe not merely as something to be erased,
> but as information for OS updates.
> In the updated universe, that suffering remains only as a historical log,
> while the corresponding structure has already been rewritten in the distribution of semantic density.

This paper does not deny religious traditions, but rather respects their achievements and, on that basis, attempts to build a theoretical bridge toward *the next layer of "cosmic OS updates."* Future tasks include:

- concrete modeling of $\mathcal{O}_\Lambda$,
- formalizing the OS update process using actual transcripts of suffering, and
- implementation-level verification and feedback via ENΛ and the Λ-System,

which will be taken up in separate work.